PRODUCT REQUIREMENTS DOCUMENT / PHASE 02

Commercial Collections

Secure, transparent B2B invoice recovery case management

DOCUMENT STATUS	OWNER / AUDIENCE
v0.1 - proposal, not approved	Prepared by Syed Nazeer | For Ricoz Innovation / project supervisor

Prepared: 08 November 2026 | Proposed market: India (B2B trade invoices)

Mandatory technology: Next.js • Node.js + Express.js • MongoDB • Git / GitHub

Scope boundary This PRD specifies a safe demonstration MVP. It does not authorize collection activity, legal communications, payment custody, or deployment with real debtor data.

What is confirmed from the assignment

- Research competing products and create a PRD before development.
- Commercial collections was described as unpaid invoices with a contemplated 15%-30% contingency fee.
- Use Next.js for frontend, Node.js + Express.js for backend, MongoDB for data, Git/GitHub for source control.
- The supervisor asked for an MVP quickly and a check-in at approximately 45% development before assigning teammates.

Unknown / unapproved: target buyer, commercial operator, case policies, final fee structure, definition of 45%, product ownership, deadline and deployment requirements.

01 Executive decision summary

Proposed product: a multi-tenant web portal where an Indian business can submit an overdue B2B invoice, track review and collection-case updates, and inspect verified recovery amounts and an explainable service-fee estimate. Internal staff review, assign and update cases. The product is a workflow tool; all real-world collection activity remains outside the demo until authorized.

Why this product?

Competitor research identifies two relevant approaches: managed recovery providers (e.g. FundRaksha and Atradius Collections) and self-service accounts-receivable software (e.g. Chaser and Kolleno). The opportunity being tested is transparent creditor-facing case tracking combined with a simple internal case desk. This is a product hypothesis, not evidence of market fit. [R1-R5]

Recommendation requested from Ricoz

- Approve a demo-only India-focused B2B trade-invoice workflow, not consumer loan collections.
- Approve three initial roles: Creditor, Case Operator and Administrator.
- Approve the P0 feature set and a testable first milestone; explicitly define what “45%” means.
- Confirm whether Ricoz operates actual collection services, supplies SaaS to agencies, or only requires an internship demonstration.

Success criterion for a first demo

End-to-end proof Using fictional data, a creditor signs in and submits an overdue invoice; an administrator accepts and assigns it; an operator logs an update; the creditor sees the activity; an admin verifies a partial receipt; the UI shows the correct outstanding balance and proposed fee - with access protections and audit history.

Document approval status

ITEM	STATUS	OWNER / NEXT ACTION
Research	Complete - competitor report v2.0	Use as evidence; no additional marketing claims assumed
PRD	Draft v0.1	Supervisor reviews assumptions and scope
Policy / pricing	Unconfirmed	Ricoz to define fees, client agreement and collection authority
Build authorization	Not granted by this document	Confirm internship terms, delivery channel and scope before real deployment

02 Objectives, users and measurable value

Problem statement

Businesses with overdue invoices can lose visibility into what happened after a claim was submitted, who owns the next action, whether payment has actually arrived, and how a success fee is calculated. Operators need structured evidence and a defensible activity record. The MVP should make these states visible without automating coercive collection work.

Target users - hypotheses to validate

PERSONA	PRIMARY JOB	MVP PERMISSION
Creditor / finance lead	Submit and monitor their company's cases	Own-company cases and documents only
Case operator	Track assigned cases and record internal activity	Assigned cases; no fee approval
Administrator	Review, assign, verify receipts and audit decisions	Authorized internal operations across case queues

North-star and supporting measures

- First-demo completion: a new claim can reach a verified receipt and fee view with no manual database edits.
- Time to first review: median interval between submission and first admin decision (measured, not guaranteed).
- Case transparency: percentage of accepted cases with an assigned owner and documented next action.
- Financial correctness: all fee, partial-payment and reversal tests pass; no unverified receipt counted as recovered.

03 Product scope and non-goals

PRIORITY	CAPABILITY	DECISION
P0 / MVP	Sign-in, company isolation, creditor case intake, evidence metadata, admin review, assignment, case timeline, verification of recorded receipts, fee calculation, basic dashboard, audit	Build
P1 / after pilot	Search/filter refinements, reminders to internal staff, exports, richer dashboard, organization invite flows, case reassignment notifications	Plan only
P2 / future	Accounting integrations, invoice bulk import, payment links, configurable workflow engines, multi-region/multi-currency, mobile app	Defer
Explicitly out	Automated debt-chasing calls/messages, legal notice generation, legal advice, coercive outreach, bank settlement, custody or transfer of funds, real public debt recovery	Do not build

Policy gate Legal notices, regulated lending recovery, debtor contact, payments and live-case processing require separately approved operating policies, security review and professional legal/compliance advice.

04 User journeys and case lifecycle

Journey A - creditor submits a case

1. Sign in to a company account.
2. Start a claim and enter debtor organization, invoice ID, issue/due dates, original value (INR), disputed flag and description.
3. Attach synthetic sample invoice/evidence in the demo and review the submitted information.
4. Submit: system validates fields, generates a case ID, captures submission time, and shows “Submitted - awaiting review”.
5. Track the case timeline, request-for-information notices and authorized progress updates.

Journey B - internal staff processes the case

6. Admin sees the intake queue and opens supporting information.
7. Admin decides “Needs information”, “Rejected”, or “Accepted”, with a written reason.
8. For accepted cases, admin sets the case fee rate and assigned operator. Rate becomes locked when work starts.
9. Operator records an internal note, due next action, and optional dispute/promise flag. This does not imply debtor contact occurred.
10. Admin records and verifies recovery evidence; results appear in balances and fee estimate.
11. Admin closes with a recorded closure reason; authorized reopenings require a rationale.

Separate workflow dimensions (avoids misleading status labels)

FIELD	ALLOWED VALUES	RULE
case.status	draft, submitted, needs_info, rejected, accepted, in_progress, on_hold, closed	Explicit transition rules; admin-owned approvals
case.disputeStatus	none, raised, resolved	Do not imply debt is undisputed
case.closureReason	paid, settled, withdrawn, uncollectible, other	Required when closing; reopening logged
recovery.status	recorded, verified, rejected, reversed	Only verified less reversals counts as recovered
promiseToPay	Activity event with due date	Never represented as collected cash

Illustrative transitions: draft -> submitted -> needs_info -> submitted; submitted -> rejected / accepted; accepted -> in_progress -> on_hold / closed; closed -> in_progress only through authorized reopen. Payment/dispute status is tracked independently.

05 Functional requirements - P0

Each row is a proposed build contract. Acceptance must be observable in the UI or API and demonstrated with test data. Changes require PRD review.

ID	REQUIREMENT	ACCEPTANCE TEST
FR-01	Secure session sign-in / sign-out	Bad credentials fail safely; authorized sessions survive refresh and expire correctly.

ID	REQUIREMENT	ACCEPTANCE TEST
FR-02	Organization-scoped data access	Creditor A cannot read, modify or download creditor B records via API or UI.
FR-03	Create/edit draft case	Required fields validated; draft persists after refresh.
FR-04	Submit invoice claim	Unique case ID created; duplicate invoice references handled explicitly; status = submitted.
FR-05	Evidence upload and metadata	Allowed types/size validated; upload requires role and company scope; file not public.
FR-06	Admin intake decisions	Accept/reject/request info changes status and stores actor, time and reason.
FR-07	Case assignment	Only admin can assign/reassign; new owner visible; changes audited.
FR-08	Timeline and internal notes	Notes/activity ordered by time with author; creditor sees only shareable events.
FR-09	Receipt recording / verification	Unverified receipt leaves recovered total unchanged; verification updates once only.
FR-10	Contingency fee and balance	Compute from net verified recoveries using approved case rate; handle partials and reversals.
FR-11	Case list and overview	Creditors see own company data; staff see authorized queues, correct status counts.
FR-12	Audit, errors, responsive UI	Critical events logged; validation useful; layout works on mobile/laptop.

06 Financial rules and example calculations

The WhatsApp brief suggested a 15%-30% contingency fee. This is an input for modeling only: the exact contracted percentage, tax treatment, inclusions, refunds, and fee-collection mechanism are NOT approved.

Proposed calculation logic

- Store monetary values as integer paise (or precise decimal). Store fee rates as integer basis points: 15%=1500, 20%=2000, 30%=3000.
- Net verified recovery = sum of approved positive recovery entries minus approved reversal entries; every adjustment is an appended event, not a silent overwrite.
- Fee estimate = net verified recovery x locked agreed rate. Round according to a policy approved before real invoicing; for the demo, round half-up to paise at aggregate level.
- Remaining principal = max(0, original invoice principal minus verified principal recovery). Interest, taxes, legal expenses, and credit notes excluded from demo calculations.
- Record who verified the receipt, when, supporting reference and whether debtor paid creditor directly. No payments are processed by the application.

Worked example - hypothetical INR values

EVENT	VERIFIED NET RECOVERY	20% FEE	REMAINING ON RS 1,00,000
Case submitted; receipt promised	Rs 0	Rs 0	Rs 1,00,000
Rs 60,000 receipt recorded, unverified	Rs 0	Rs 0	Rs 1,00,000
Rs 60,000 verified	Rs 60,000	Rs 12,000	Rs 40,000
Rs 10,000 correction/reversal verified	Rs 50,000	Rs 10,000	Rs 50,000

Truthfulness requirement The interface must say “Recorded, awaiting verification” until evidence is approved. Never label promised or unverified funds “Recovered”. The 20% rate above is illustrative, not Ricoz pricing.

Financial edge cases required in tests

- Duplicate receipt reference: second verification must be rejected / idempotent.
- Partial receipts: fee based on verified portion, not the full invoice.
- Reversal: create a linked negative adjustment with approval; do not delete the original receipt.
- Negative, zero, excessive, foreign-currency or malformed amounts: reject or require explicit approval by policy.
- Rate changes after work starts: prohibit by default; any policy exception requires admin authorization and recorded rationale.
- Over-recovery, settlement write-offs and interest: mark out of MVP until rules are approved.

07 Screen and interaction inventory

SCREEN	ROLE	ESSENTIAL ELEMENTS
Login / demo access	All	Safe sign-in, errors, session status.
Creditor overview	Creditor	Submitted / in-progress / awaiting-info cases; verified recovery, next actions.
New case / case detail	Creditor	Invoice form, proof, review, timeline, owner, fee preview.
Admin review queue	Admin	Pending claims, evidence, decisions, reasons.
Operator worklist	Operator	Assigned cases, due next actions, filtered internal notes.
Recovery verification	Admin	Recorded receipts, evidence, approve/reject/reverse.
Audit view	Admin	Immutable event timeline and filter by actor/time/case.

Interface principles

- Light, readable and professional visual system; no unnecessary dashboard widgets or duplicate actions.
- Persistent navigation, visible case stage, clearly differentiated unverified vs verified funds.
- Accessible form labels, keyboard support, visible focus and mobile-friendly tables.
- Confirmation before destructive changes; clear retry states and no false success banners.

08 Architecture and technical constraints

The company-mandated stack is authoritative: Next.js frontend, separate Node.js + Express.js API, MongoDB persistence, Git and GitHub. The choices below are implementation proposals that can be reviewed with the team.

Logical architecture

Request flow Next.js web UI -> authenticated HTTPS calls -> Express REST API -> MongoDB data collections; file attachments use private object storage with authorized retrieval. Never give the browser database credentials.

LAYER	PROPOSED DESIGN	RESPONSIBILITY
Web app	Next.js + TypeScript (proposal)	UI/routes, form validation, accessible layout, frontend session handling.
API	Node.js + Express + TypeScript (proposal)	Validation, authorization, workflow, idempotency, fee math, audit.
Database	MongoDB	Organizations, users, debtors, invoices, cases, receipts and audit.
File storage	Private object store, provider TBD	Upload limits, malware scanning policy, signed access or proxy download.
Delivery	GitHub + isolated demo deployment	PR checks, separate environment variables, synthetic seeded demo records.

Proposed MongoDB collections

COLLECTION	IMPORTANT FIELDS / RULES
organizations	_id, name, country, status; creditor company tenant.
users	_id, orgId (for creditor), role, email, passwordHash/providerId, active.
debtors	_id, creditorOrgId, registeredName, contact metadata - keep personal data minimal.
invoices	_id, creditorOrgId, debtorId, invoiceNo, principalPaise, currency=INR, issueDate, dueDate.
cases	_id, creditorOrgId, invoiceId, status, disputeStatus, operatorId, feeRateBps, createdAt.
documents	_id, creditorOrgId, caseId, category, blobRef, mime, checksum, uploadedBy.
activities	_id, caseId, creditorOrgId, actorId, kind, visibility, occurredAt, text.
recoveries	_id, caseId, creditorOrgId, amountPaise, status, externalRef, verifiedBy, reversedFrom.
auditEvents	_id, caseId, actorId, orgId, eventType, timestamp, non-secret diff.

Index proposals: unique (creditorOrgId, invoiceNo) if business rules allow; (creditorOrgId, status, createdAt); (operatorId, status); unique recovery idempotency key per case. Avoid saving uploaded files directly inside MongoDB documents.

Suggested Express API groups

ENDPOINT GROUP	EXAMPLE ROUTES (DRAFT)
/api/v1/auth	POST /login, POST /logout, GET /me
/api/v1/cases	GET /, POST /, GET /:id, PATCH /:id, POST /:id/submit
/api/v1/cases/:id/review	POST /decision, POST /assign, POST /activity
/api/v1/cases/:id/documents	POST /, GET /:documentId (authorized access)
/api/v1/cases/:id/recoveries	POST /record, POST /:receiptId/verify, POST /:receiptId/reverse
/api/v1/dashboard	GET /summary; role-scoped calculated totals

09 Security, privacy and operational guardrails

- Tenant isolation must be enforced inside Express queries and document endpoints; never trust a client-supplied organization ID.
- Access is deny-by-default: creditor (own company), operator (assigned cases), admin (authorized work queue); separate private staff notes from client-facing updates.
- Use secure server-side sessions or appropriately protected tokens, TLS, httpOnly cookies where applicable, CSRF protection for cookie sessions, rate limiting and password hashing.
- Validate requests on the server. Restrict uploads by MIME, extension and size; scan prior to production use and keep storage private.
- Security-relevant changes are audited with minimal redaction-safe metadata. Do not log passwords, bank details, documents or access tokens.
- Use fictional debtor contacts and invoices in demos. Production handling of personal data and retention requires a documented privacy/compliance decision.
- No automated intimidation, unauthorized third-party messaging, legal notices, disbursements or payment capture in the demo.

Compliance notes: India’s DPDP Act / Rules have a phased enforcement framework; design for data minimization, purpose clarity and security, then obtain legal advice on applicability and implementation dates. The MSME Samadhaan public remedy may apply to eligible micro/small suppliers but is not implemented in this MVP. OWASP ASVS provides relevant security requirements. [R6-R8]

10 Non-functional requirements and quality gates

ID	REQUIREMENT	VERIFY
NFR-01	Usable on 360px mobile viewport through standard laptop/desktop sizes.	Manual responsive tests at 360, 768, 1280px; no horizontal overflow in main tasks.
NFR-02	Accessible semantic forms and keyboard navigation.	Keyboard walkthrough, focus indicators, labels and error associations.
NFR-03	API errors are actionable but do not disclose internals.	Invalid requests return consistent 4xx; unhandled errors generic 5xx; no stack traces.
NFR-04	Consistent data after reload and duplicate submissions.	Refresh/simulated retries; no duplicate receipt verification or cases.
NFR-05	Demo environment isolates configuration and secrets.	No .env secrets in Git; fresh install runs with documented configuration.
NFR-06	Core workflows are covered by automated tests.	Unit: rates/reversals; integration: roles/tenants; E2E: case creation/review.
NFR-07	Activity events attributable and tamper-resistant.	Audit actor, timestamp, event; unauthorized edit/delete rejected.
NFR-08	Initial performance measured, not promised.	Record p95 API latency and page load in pilot; targets set after baseline.

Definition of done - every P0 feature

- Working frontend flow and authorized backend endpoint connected to MongoDB.
- Server-side validation, permission checks and meaningful error states present.
- At least one happy-path test plus negative / unauthorized scenario covered.
- Demo data persists across refresh; audit record exists for relevant changes.
- README updated with setup, feature status, screenshots and reproducible test steps.

11 Milestones, 45% checkpoint and handoff

Rajesh asked to reconnect after 45% development. No approved denominator or acceptance standard exists. The following is a proposed feature-based checkpoint and must not be described as precisely 45% without supervisor agreement.

STAGE	DELIVERABLE	GATE
Gate A - PRD sign-off	Competitor report + PRD v0.1, decision answers, prioritized backlog	Supervisor confirms business model and scope.
Gate B - foundation	Repo, Next/Express/Mongo connection, sign-in, roles and tenant isolation	Login works; cross-tenant access tests fail safely.
Proposed 45% check-in	Creditor intake, evidence metadata, admin queue + decision, basic timeline	Demo end-to-end submit -> review -> update; GitHub link and checklist.
Gate C - full MVP	Operator tasks, verification/reversals, fee calculations, dashboard, audit	Complete financial tests; no false success counts.

STAGE	DELIVERABLE	GATE
Gate D - handoff	Hosted demo if approved, code, README, synthetic data, test evidence	Ricoz agrees destination, repository ownership and maintenance owner.

Suggested 45% demonstration checklist (provisional)

- [ ] Frontend and Express API run using documented commands; MongoDB demo connection works.
- [ ] Two creditor organizations exist with synthetic invoices; no cross-tenant visibility.
- [ ] Creditor can draft, submit and view own claim; evidence is referenced securely.
- [ ] Admin can accept, reject, or request information with recorded rationale.
- [ ] Assigned case and creditor-visible timeline update without page reload / fake success.
- [ ] GitHub contains README, sample environment config, tests, screenshots and known limitations.

The first checkpoint intentionally excludes fee logic / verified recovery unless the supervisor prioritizes it differently. The milestone demonstrates a meaningful vertical slice, not just polished screens.

12 Delivery, demonstration and review process

12. Share PRD PDF and editable Word version with Rajesh on WhatsApp and request explicit acceptance or changes.
13. Confirm GitHub repository ownership, whether code should stay private, and who will review pull requests.
14. Use a non-production, isolated demo deployment and fictional data; do not share permanent production credentials by chat.
15. At each milestone, share: live demo URL (if approved), repository link, short feature checklist, test outcomes and unresolved decisions.
16. Do not label an unreviewed PRD “approved” or the demo “production-ready.”

13 Risks, dependencies and key decisions

RISK / DEPENDENCY	CONSEQUENCE	PROPOSED MITIGATION
Business model undefined	Wrong roles, workflows and value proposition	Approve managed-service vs agency SaaS vs prototype.
45% unspecified	Misunderstood milestone / rework	Approve feature-based checkpoint and sign-off owner.
Unclear authority for outreach	Compliance and reputational risk	Demo only; no outbound debtor contact or legal automation.
Incorrect financial model	Misleading balances, wrong fee estimates	Verified-only ledger, idempotency and reversal tests.
Shared organization data	Confidential information exposed	Tenant-scoped backend queries + negative tests.
Unclear internship / ownership	Disputes over deliverables and responsibility	Confirm written terms, repo ownership and expectations.
Unfunded hosting / storage	Demo fails after free quotas	Choose hosting limits and retention before deployment.

Approval questions for Rajesh (answer before implementation)

- Q1. Is this a Ricoz-managed debt recovery service, SaaS for existing agencies, or a demonstration product?
- Q2. Is the target India B2B trade invoices only? Are consumer loans and lending clients excluded?
- Q3. Who is authorized to accept cases, communicate with debtors and verify financial recovery?
- Q4. Which three functions must be in the first prototype: intake, staff case management, payment/fee records, or something else?
- Q5. Is 15%-30% a fixed pricing proposal, negotiated per case, or just an illustrative range?
- Q6. What exactly counts toward the “45%” milestone, and who will approve it?
- Q7. What is the PRD review deadline and required submission channel (WhatsApp, dashboard, GitHub, email)?
- Q8. Who owns the repository and resulting code, who supervises it, and what are the agreed internship terms?

Decision register

DECISION ID	PROPOSED DECISION	STATUS
D-01	India B2B invoice demonstration, not real operations	Awaiting Ricoz
D-02	Creditor + operator + admin roles	Awaiting Ricoz
D-03	Partial/verified recovery tracked; no funds transferred	Awaiting Ricoz
D-04	No automated debtor communications or legal actions	Recommended safety boundary
D-05	First checkpoint defined by working case intake + review	Awaiting Ricoz
D-06	Rate and tax policy signed off before financial production use	Awaiting Ricoz

14 Research sources and evidence notes

Sources underpinning this PRD were checked against public vendor / official pages. Product features and pricing are vendors' statements; business model, priorities, technical design, and acceptance criteria are this document's proposals. Review date: 08 November 2026.

[R1] Commercial Collections - Competitor Research v2.0 | Companion research PDF prepared 07 Nov 2026

[R2] FundRaksha - commercial debt recovery | https://www.fundraksha.com/services/commercial-debt-recovery

[R3] FundRaksha - success-fee pricing | https://legal.fundraksha.com/pricing

[R4] CARMA Recover - managed collections | https://www.carmaone.ai/carma-recover

[R5] Atradius Collections - online collection | https://atradiuscollections.com/global/services/online-debt-collection

[R6] Ministry of MSME - MSME Samadhaan | https://ramp.msme.gov.in/ramp/RAMP-initiative/msme-samadhaan/msme-samadhaan

[R7] Ministry of Electronics and IT - DPDP Rules materials | https://www.meity.gov.in/documents/act-and-policies/digital-personal-data-protection-rules-2025-gDOxUjMtQWa?pageTitle=Digital-Personal-Data-Protection-Rules-2025-2

[R8] OWASP - ASVS 5.0 | https://asvs.owasp.org/

Review status This is PRD v0.1: detailed enough to estimate and prototype, but not approved for real-life collections, production deployment or independent transfer of funds. Obtain explicit sign-off on decisions D-01 to D-06.